Project: Worked examples
Title: Press and shrink fit
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

A shaft (solid or hollow) pressed or shrunk into a hub: the interface pressure for a given diametral interference, the hoop and radial stresses at the interface in hub and shaft, von Mises utilisation against each yield strength, the axial assembly force and the torque the joint can carry by friction, and the temperature rise to shrink-fit the hub.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Radial interference | $\delta_r = \delta/2$ |
| Interface pressure, the hub and the shaft as thick cylinders | $p = \delta_r / [\ (R/E_h)\cdot((r_o^2 + R^2)/(r_o^2 - R^2) + \nu_h) + (R/E_s)\cdot((R^2 + r_i^2)/(R^2 - r_i^2) - \nu_s)\ ]$ |
| Stresses at the hub bore | $\sigma_\theta = p\cdot(r_o^2 + R^2)/(r_o^2 - R^2)$; $\sigma_r = -p$ |
| Stresses at the shaft surface | $\sigma_\theta = -p\cdot(R^2 + r_i^2)/(R^2 - r_i^2)$; $\sigma_r = -p$ |
| Axial force and torque the fit holds by friction | $F = \mu\cdot p\cdot 2\pi R\cdot L$; $T = \mu\cdot p\cdot 2\pi R^2\cdot L$ |
| Temperature difference to assemble by heating the hub, with assembly clearance $c_a$ | $\Delta T = (\delta + c_a)/(\alpha\cdot 2R)$ |

Basis: Press and shrink fits; stresses in pressurised cylinders[1](§3-16, §3-17); Thick-walled cylinders under pressure[2](Ch. 13, Table 13.5).

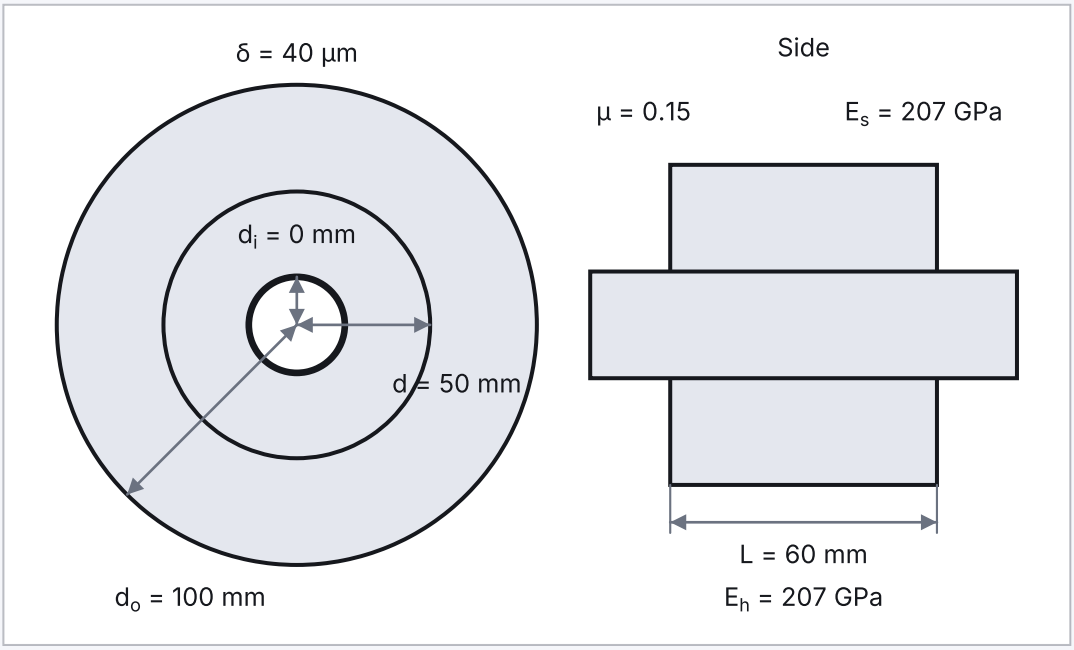



Figure 1 — A hub shrunk onto a shaft: section across the interface, and the engagement length beside it. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| d | Interface (nominal) diameter | 50 | mm | drawing: DRG-HUB-04 rev C | 1 … 1000 mm |
| δ | Diametral interference | 40 | µm | drawing: DRG-HUB-04 (H7/p6 mean) rev C | 1 … 500 µm |
| $d_o$ | Hub outer diameter | 100 | mm | drawing: DRG-HUB-04 rev C | 1 … 2000 mm |
| $d_i$ | Shaft bore diameter (0 for a solid shaft) | 0 | mm | drawing: DRG-SHAFT-12 rev A | 0 … 1000 mm |
| L | Engagement length | 60 | mm | drawing: DRG-HUB-04 rev C | 1 … 2000 mm |
| $E_h$ | Hub Young's modulus | 207 | GPa | spec: EN 10025 S355, E per textbook | 10 … 450 GPa |
| $\nu_h$ | Hub Poisson's ratio | 0.292 | | spec: steel | 0.1 … 0.45 |
| $E_s$ | Shaft Young's modulus | 207 | GPa | spec: steel | 10 … 450 GPa |
| $\nu_s$ | Shaft Poisson's ratio | 0.292 | | spec: steel | 0.1 … 0.45 |
| µ | Coefficient of friction at the interface | 0.15 | | assumption: Dry steel on steel, pressed; Shigley Table 7-? range 0.1–0.3. | 0.05 … 0.4 |
| $S_{y,h}$ | Hub yield strength | 355 | MPa | spec: EN 10025 S355 | 50 … 2000 MPa |
| $S_{y,s}$ | Shaft yield strength | 355 | MPa | spec: EN 10025 S355 | 50 … 2000 MPa |
| $\alpha_h$ | Hub thermal expansion coefficient | 0.000012 | 1/K | spec: steel | 0.000005 … 0.00003 1/K |
| $c_a$ | Assembly clearance wanted for shrink fitting | 20 | µm | assumption: Assembly clearance for a shrink fit. | 0 … 500 µm |

## 3. Assumptions

A1. Both parts are long cylinders in plane stress; the interface pressure is uniform along the engagement. (Lamé thick-cylinder relations; end effects neglected.)
A2. The friction coefficient is as entered and the same for assembly and for torque transmission. (Enter it with its source; it dominates the force and torque results.)
A3. Yield is checked by the von Mises equivalent stress at the hub bore, at the shaft surface and, for a hollow shaft, at the shaft bore (where the hoop stress is largest: $2 \cdot p \cdot R^2/(R^2 - r_i^2)$, with no radial stress), the larger of the two shaft values governing. (Axial stress taken as zero.)
A4. µ (Coefficient of friction at the interface) is taken as 0.15. (Dry steel on steel, pressed; Shigley Table 7-? range 0.1–0.3.)
A5. $c_a$ (Assembly clearance wanted for shrink fitting) is taken as 20 µm. (Assembly clearance for a shrink fit.)

## 4. Working

[15] $R = d / 2$
$= 50\ \text{mm} / 2$
$= 25\ \text{mm}$
Interface radius

[16] $r_o = d_o / 2$
$= 100\ \text{mm} / 2$
$= 50\ \text{mm}$
Hub outer radius

[17] $r_i = d_i / 2$
$= 0\ \text{mm} / 2$
$= 0\ \text{mm}$
Shaft bore radius

[18] $\delta_r = \delta / 2$
$= 40\ \mu\text{m} / 2$
$= 20\ \mu\text{m}$
Radial interference

[19] $C_h = (R / E_h) * ((r_o^2 + R^2) / (r_o^2 - R^2) + \nu_h)$ [1](§3-16)
$= 25\ \text{mm} / 207\ \text{GPa} \times (((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2) + 0.292)$
$= 0.23655\ \text{mm/GPa}$
Hub compliance

[20] $C_s = (R / E_s) * ((R^2 + r_i^2) / (R^2 - r_i^2) - \nu_s)$ [1](§3-16)
$= 25\ \text{mm} / 207\ \text{GPa} \times (((25\ \text{mm})^2 + (0\ \text{mm})^2) / ((25\ \text{mm})^2 - (0\ \text{mm})^2) - 0.292)$
$= 0.085507\ \text{mm/GPa}$
Shaft compliance

[21] $p = \delta_r / (C_h + C_s)$ [1](eq. 3-56)
$= 20\ \mu\text{m} / (0.23655\ \text{mm/GPa} + 0.085507\ \text{mm/GPa})$
$= 62.1\ \text{MPa}$
Interface pressure

[22] $\sigma_{t,hub} = p * (r_o^2 + R^2) / (r_o^2 - R^2)$ [1](eq. 3-57)
$= 62.1\ \text{MPa} \times ((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2)$
$= 103.5\ \text{MPa}$
Hub hoop stress at the bore

[23] $\sigma_{r,hub} = -p$
$= -62.1\ \text{MPa}$
$= -62.1\ \text{MPa}$
Hub radial stress at the bore

[24] $\sigma_{t,shaft} = -p * (R^2 + r_i^2) / (R^2 - r_i^2)$ [1](eq. 3-58)
$= -62.1\ \text{MPa} \times ((25\ \text{mm})^2 + (0\ \text{mm})^2) / ((25\ \text{mm})^2 - (0\ \text{mm})^2)$
$= -62.1\ \text{MPa}$
Shaft hoop stress at the surface

[25] $\sigma_{r,shaft} = -p$
$= -62.1\ \text{MPa}$
$= -62.1\ \text{MPa}$
Shaft radial stress at the surface

[26] $\sigma_{vm,hub} = \text{sqrt}(\sigma_{t,hub}^2 - \sigma_{t,hub} * \sigma_{r,hub} + \sigma_{r,hub}^2)$
$= \text{sqrt}((103.5\ \text{MPa})^2 - 103.5\ \text{MPa} \times (-62.1\ \text{MPa}) + (-62.1\ \text{MPa})^2)$
$= 144.9\ \text{MPa}$
Hub von Mises stress at the bore

[27] $\sigma_{vm,shaft} = \text{sqrt}(\sigma_{t,shaft}^2 - \sigma_{t,shaft} * \sigma_{r,shaft} + \sigma_{r,shaft}^2)$
$= \text{sqrt}((-62.1\ \text{MPa})^2 - (-62.1\ \text{MPa}) \times (-62.1\ \text{MPa}) + (-62.1\ \text{MPa})^2)$
$= 62.1\ \text{MPa}$
Shaft von Mises stress at the surface

[28] $\sigma_{vm,shaft,bore} = \sigma_{vm,shaft}$
= 62.1 MPa
= 62.1 MPa
Shaft von Mises stress at the centre (solid shaft: equal hoop and radial stress, the same as at the surface)
[29] $\sigma_{vm,shaft,max} = \max(\sigma_{vm,shaft}, \sigma_{vm,shaft,bore})$
= max(62.1 MPa, 62.1 MPa)
= 62.1 MPa
Largest shaft von Mises stress, surface or bore
[30] $F_a$ = μ * p * 2 * pi * R * L
= 0.15 × 62.1 MPa × 2 × π × 25 mm × 60 mm
= 87.792 kN
Axial assembly (press) force
[31] T = μ * p * 2 * pi * R^2 * L
= 0.15 × 62.1 MPa × 2 × π × (25 mm)^2 × 60 mm
= 2194.8 N·m
Torque capacity by friction
[32] dT = (δ + $c_a$) / ($\alpha_h$ * d)
= (40 μm + 20 μm) / (0.000012 1/K × 50 mm)
= 100 K
Hub temperature rise for shrink fitting

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| p | 62.1 MPa | Interface pressure |
| $\sigma_{t,hub}$ | 103.5 MPa | Hub hoop stress at bore |
| $\sigma_{t,shaft}$ | -62.1 MPa | Shaft hoop stress at surface |
| $\sigma_{vm,hub}$ | 144.9 MPa | Hub von Mises stress |
| $\sigma_{vm,shaft}$ | 62.1 MPa | Shaft von Mises stress at the surface |
| $\sigma_{vm,shaft,bore}$ | 62.1 MPa | Shaft von Mises stress at the shaft bore |
| $\sigma_{vm,shaft,max}$ | 62.1 MPa | Largest shaft von Mises stress (surface or bore) |
| $F_a$ | 87.792 kN | Assembly force |
| T | 2194.8 N·m | Torque capacity |
| dT | 100 K | Shrink-fit temperature rise |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Hub yield | 144.9 MPa | 355 MPa | 0.408 | |
| PASS | Shaft yield, surface and bore | 62.1 MPa | 355 MPa | 0.175 | |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.press-fit@1.1.0 Content hash ab00ab9413e55979 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.